SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	Kanish Kumar V	**Reg. No.:**	192524361
Section / Batch:	B	**Date:**	13/07/2026

Code of Conduct

I Kanish Kumar/(192524361) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kanish Kumar V

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
	GRAND TOTAL:	**100 Marks**	


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CALCULATOR APPLICATION – SUITABLE DATA STRUCTURE FOR ARITHMETIC EXPRESSION EVALUATION
CSA0309-DATA STRUCTURES
Done by
Kanish Kumar V
(192524361)
Supervised By
Dr Uma Priyadarshini
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CALCULATOR APPLICATION – SUITABLE DATA STRUCTURE FOR ARITHMETIC EXPRESSION EVALUATION

CSA0309-DATA STRUCTURES

Done by
Kanish Kumar V
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ABSTRACT

- Arithmetic expression evaluation is a fundamental function in calculator applications.
- Choosing the right **data structure** improves processing speed, accuracy, and memory efficiency.
- **Stacks** are the most suitable data structure because they follow the **Last-In, First-Out (LIFO)** principle.
- Stacks efficiently manage **operators, operands, and parentheses** during expression evaluation.
- Overall, stacks provide a reliable, efficient, and scalable solution for arithmetic expression processing.